higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

LOGIC SYSTEMS N5

26 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

A	B	C	J_A	K_A	J_B	K_B	J_C	K_C	
0	1	0	0	Ø	Ø	0	1	Ø	
0	1	1	1	Ø	Ø	1	Ø	1	
1	0	0	Ø	0	0	Ø	1	Ø	Λ
1	0	1	Ø	0	1	Ø	Ø	1	
1	1	0	Ø	0	Ø	0	1	Ø	
1	1	1	Ø	1	Ø	0	Ø	1	Λ
0	0	0	0	Ø	1	Ø	0	Ø	
0	0	1	0	Ø	1	Ø	Ø	1	

$Q_n \rightarrow Q_n + 1$	J	K
0 → 0	0	Ø
0 → 1	1	Ø
1 → 0	Ø	1
1 → 1	Ø	0

J_A = BC Λ

	00	01	11	10	
0	0	0	Ø	Ø	
1	0	1	Ø	Ø	Λ

K_A = BC Λ

	00	01	11	10	
0	Ø	Ø	0	0	
1	Ø	Ø	1	0	Λ

J_B = 'A+C Λ

	00	01	11	10	
0	1	Ø	Ø	0	
1	1	Ø	Ø	1	Λ

K_B = 'AC Λ

	00	01	11	10	
0	Ø	0	0	Ø	
1	Ø	1	0	Ø	Λ

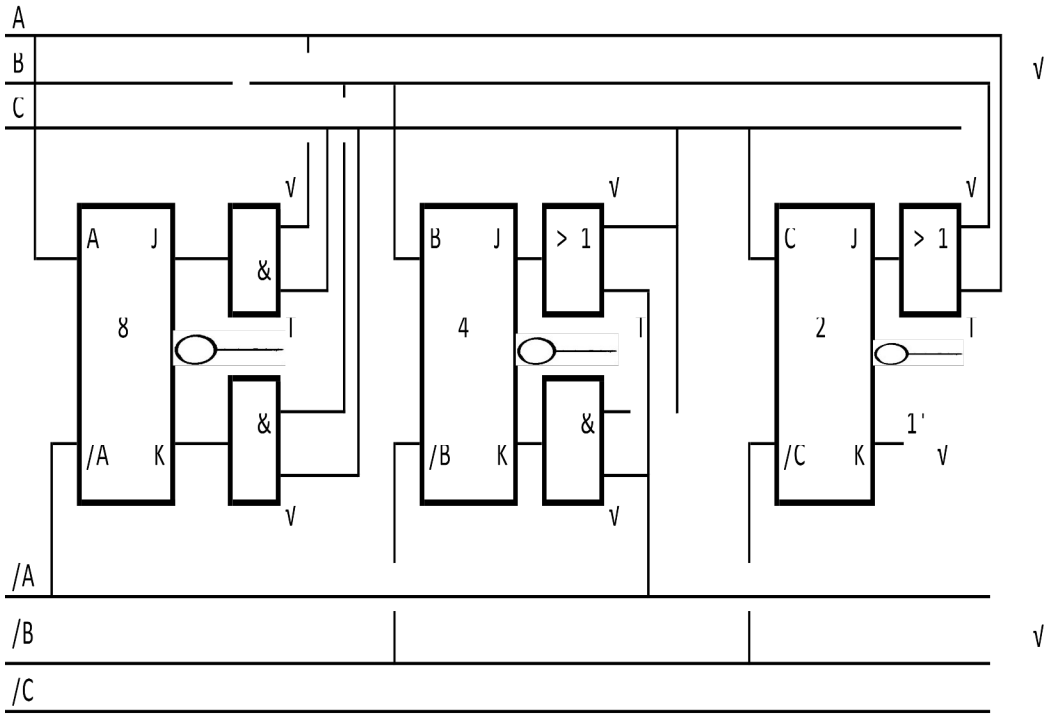
J_C = B + A Λ

	00	01	11	10	
0	0	1	1	1	
1	Ø	Ø	Ø	Ø	Λ

K_C = 1 Λ

	00	01	11	10	
0	Ø	Ø	Ø	Ø	
1	1	1	1	1	Λ

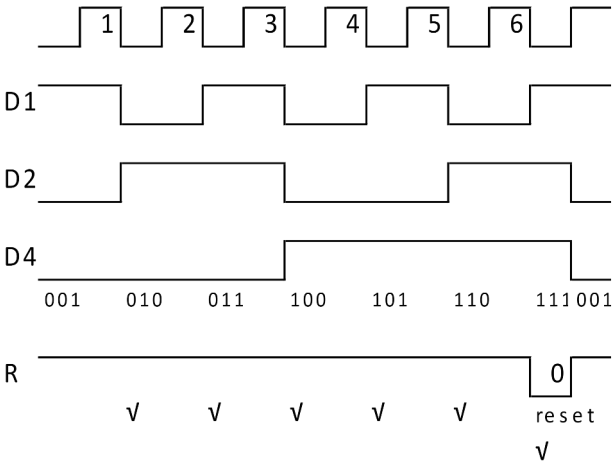
(7)



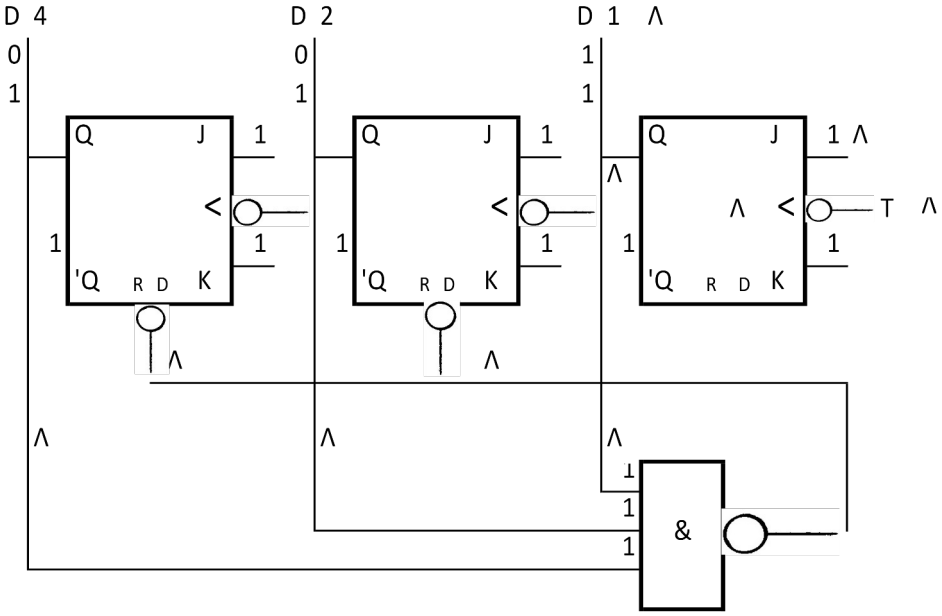
(8)
[15]

QUESTION 2

	D4	D2	D1	T	
	0	0	1	0	T
	0	1	0	1	√
	0	1	1	2	
	1	0	0	3	√
	1	0	1	4	
	1	1	0	5	√
	1	1	1	6	
Reset	0	0	1	6	√



(10)



(Λ = ½ mark)
(10 × ½) (5)
[15]

QUESTION 3

3.1 True
3.2 False
3.3 False
3.4 True
3.5 False
3.6 True
3.7 False
3.8 True
3.9 False
3.10 False
3.11 True
3.12 True
3.13 True

(13 × 1) **[13]**

QUESTION 4

4.1 Right-shift Johnson counter (1)

4.2 Left-shift Johnson counter (1)

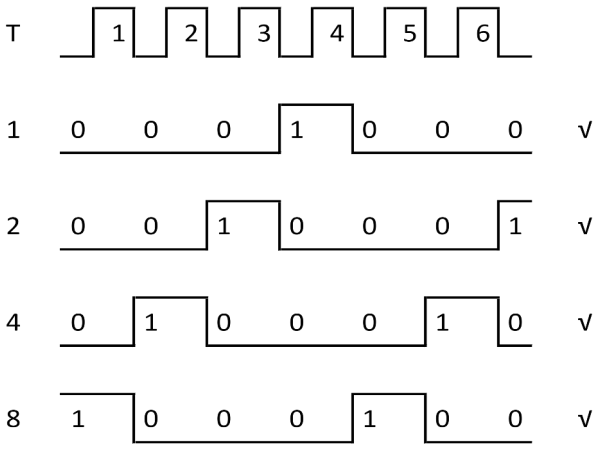
4.3 Left-shift register (1)

4.4 Right-shift ring counter (1)

4.5 Left-shift ring counter (1)

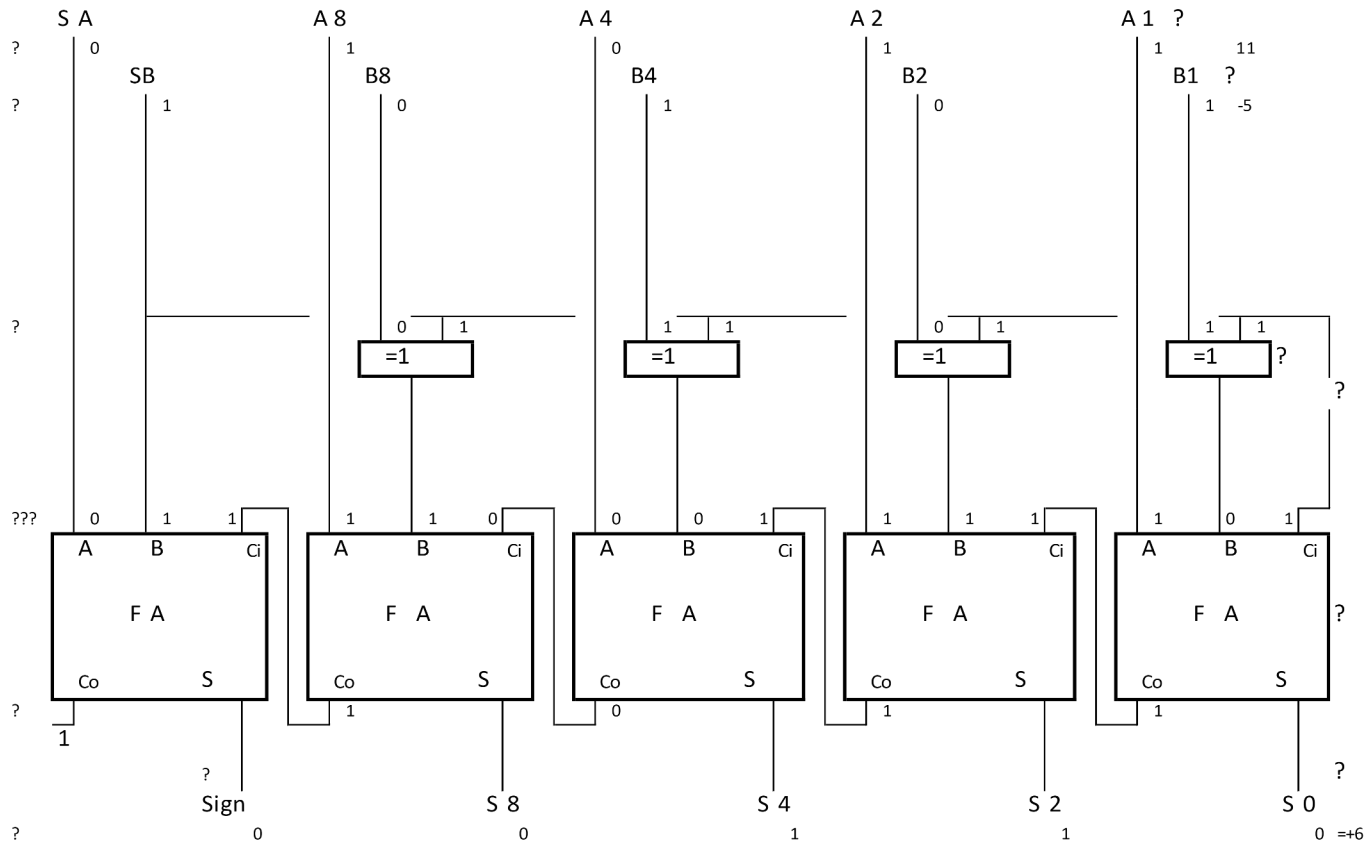
4.6 0100, 0010, 0001 (3)

4.7



(4)
[12]

QUESTION 5



(Logic levels) (8)
(Circuit) (7)
[15]

QUESTION 6

6.1 6.1.1 Dynamic MOS RAM cell
6.1.2 Static MOS RAM cell
6.1.3 Static bipolar RAM cell

(3 × 1) (3)

6.2 6.2.1 B, C (2)
6.2.2 A (1)
6.2.3 A, B, C (3)
6.2.4 None (1)
6.2.5 A, B (2)
6.2.6 A (1)
6.2.7 B, C (2)
[15]

QUESTION 7

Explain any THREE of the following printers:

7.1 LASER PRINTER

- A laser beam projects an image on the page to be printed onto an electrically-charged rotating drum coated with selenium.
- Photoconductivity removes charge from the areas exposed to light.
- Dry ink (toner) particles are then electrostatically picked up by the drum's charged areas.
- The drum then prints the image onto paper by
- direct contact and heat, which fuses the ink to the paper.

7.2 A DOT-MATRIX PRINTER

- has a print head that runs back and forth, or in an up and down motion
- on the page and prints by impact, striking an ink-soaked cloth ribbon against the paper
- much like the print mechanism on a typewriter.
- However, unlike a typewriter or daisy-wheel printer
- letters are drawn out of a dot matrix and thus varied fonts and arbitrary graphics can be produced.

7.3 INK-JET PRINTER

- These printers produce a fine, high-velocity stream of ink
- which is deflected vertically and horizontally electrically
- to form characters on the paper.
- The stream is deflected generally by passing the ink stream
- through an electrostatic field such as that used in an oscilloscope.

7.4 CHAIN PRINTER

- Paper moves vertically in front of a moving chain
- consisting of raised characters.
- When the desired character passes its print position
- the hammer in that position is driven against the ink ribbon
- and the paper so that the character is printed on the paper.

(Any 3 × 5) **[15]**

TOTAL: 100

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